CHEST

Extremes of Age Decrease Survival in Adults After Lung Transplant

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PODCAST

BACKGROUND: Age has been implicated as a factor in the plateau of long-term survival after lung transplant.

METHODS: We used data from the Scientific Registry of Transplant Recipients to identify all recipients of lung transplant aged ≥18 years of age between January 1, 2006, and February 19, 2015. A total of 14,253 patients were included in the analysis. Survival was estimated using a nonproportional hazard model and random-survival forest methodology was used to examine risk factors for death. Final selection of model variables was performed using bootstrap aggregation. Age was analyzed as both a continuous and categorical variable (age <30, 30-55, and >55 years). Risk factors for death were obtained for the entire cohort and additional age-specific risk factors were identified for each age category.

RESULTS: The median age at transplant was 59 years. There were 1,098 (7.7%) recipients <30 years, 4,201 (29.5%) 30 to 55 years, and 8,954 (62.8%) >55 years of age. Age was the most significant risk factor for death at all time-points following transplant and its impact becomes more prominent as time from transplant increases. Risk factors for death for all patients included extremes of age, higher creatinine, single lung transplant, hospitalization before transplant, and increased bilirubin. Risk factors for death differed by age with social determinants of health disproportionately affecting survival for those in the youngest age category.

CONCLUSIONS: The youngest and oldest adult recipients experienced the lowest posttransplant survival through divergent pathways that may present opportunities for intervention to improve survival after lung transplant. CHEST 2020; 157(4):907-915

KEY WORDS: lung transplant; lung transplant outcomes; organ allocation; random survival forest

FOR EDITORIAL COMMENT, SEE PAGE 757

ABBREVIATIONS: CF = cystic fibrosis; ISHLT = International Society of Heart and Lung Transplantation; LAS = lung allocation score; RSF = random survival forest; SRTR = Scientific Registry of Transplant Recipients; VIMP = variable importance measure

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Despite scientific advances and improvements in post-transplant care, long-term survival in lung transplantation has plateaued in the past decade with a 5-year survival rate of 55.6% [1]. A possible explanation for this plateau is that age of lung transplant candidates has increased in the past decade with those aged ≥65 years comprising 14.6% of recipients in 2008 and 30.1% in 2017, a trend largely attributed to the implementation of the Lung Allocation Score (LAS) system.[1] Multiple US and international studies have demonstrated increased post-transplant mortality among older lung transplant recipients,[2-5] although others have not found this.[6,7] Generalizability of prior analyses has been limited by center-specific practice variation, variable age classification, and measurements spanning changes in lung allocation systems.

Disparate effects of age on survival have been reported in US analyses before implementation of the LAS in 2005 and beyond.[5,8,9] The purpose of this study was to characterize the association of long-term survival following lung transplant with age as a continuous variable to provide a more robust assessment, testing for a nonlinear relationship between age and mortality. This was performed in a modern cohort of patients to definitively address the effect of advancing age among US transplant recipients. We used a novel methodology; random survival forest (RSF) modeling, a nonparametric statistical technique that allows this study to overcome limitations of prior studies to study the effect of age. Our hypothesis was that increased age is associated with decreased long-term survival defined as greater than 5 years after lung transplant.

Methods

Population

The study used data from the Scientific Registry of Transplant Recipients (SRTR), which includes data on every US organ donor, waitlist candidate, and transplant recipient that is submitted by members of Organ Procurement and Transplantation Network.[10] This is a retrospective cohort study of adult (age ≥18 years) lung transplant recipients between January 1, 2006, and February 19, 2015, representing a contemporary patient cohort in the LAS era. January 1, 2006, was chosen to allow for the washout period of May 2005-January 2006 in which characteristics of the candidates were mixed and may have reflected the prior wait time–based system. February 15, 2015, was chosen as the end date because the revised LAS calculation was implemented on that date making future LAS values after this revision not comparable to prior values. Multiorgan transplants were excluded. The study was approved by the Cleveland Clinic institutional review board (16-1639).

Variables

Multivariable modeling included recipient and donor characteristics, and paired variables used to match the donor with the recipient. LAS at listing and last calculated LAS before transplant were recorded.[11] The LAS is a normalized score ranging from 0 to 100 that is used to prioritize lung waitlist candidates.[12] For the LAS, pulmonary diagnoses are categorized into four groups on the basis of survival probability and pathophysiology of the underlying disease. Diagnosis groups include group A, obstructive lung disease; group B, pulmonary vascular disease; group C, cystic fibrosis (CF) and immunodeficiency disorders; and group D, restrictive lung disease.

Analysis

Analysis was performed using SAS statistical software (SAS v9.1, SAS Institute Inc.) and R 3.0.1 package.[13] Continuous variables were summarized using means and standard deviations and categorical variables by frequencies and percentages. The Wilcoxon rank sum test was used to compare continuous variables; χ^2 or Fisher exact test was used for categorical variables. Kaplan-Meier,[14] nonproportional hazard, and nonparametric RSF analyses[15,16] were used to analyze association of survival after transplant with recipient age. Death data were obtained from the SRTR database, which is linked with the Social Security Death Master File; causes of death are listed in e-Table 1. Patients were followed to time of death or the end of follow-up and censored at retransplant.

A parametric multiphase nonproportional hazard model was used because of the presence of two distinct hazard phases with different risk factors: the early phase, which predominated before 2.5 years posttransplant, and the late phase, which outweighed risk factors from the early phase after 2.5 years.[17] Bootstrap aggregation was used to rank variables according to reliability of statistical association using variables listed in e-Appendix 1. One thousand bootstrap samples were analyzed with a P value criterion of .05 by automated stepwise regression and the proportion of times a variable was selected as the measure of reliability.

As a complementary analysis, RSF was used to validate parametric findings while avoiding statistical assumptions of a parametric model, maximize sensitivity to variable interactions, and assess linear and nonlinear relationships with survival.[15] Forest computations were implemented using open source randomForestSRC R-software; 96 variables were used to grow 500 trees.[18] RSF uses a variable importance measure (VIMP) to rank effect of variables on outcomes, with a larger VIMP value indicating greater effect.[19] To address possible confounders, VIMP interactions between individual variables and age were analyzed (e-Appendix 2).

Age was first assessed as a continuous variable and then as a categorical variable with subgroups of <30, 30 to 55, and >55 years of age in multivariable mortality analysis. These break-points were chosen on the basis of divergent survival patterns seen in the survival model with survival trajectory changing at 30 and 55 years, respectively. This strategy was used to group individuals on the basis of similar slopes for risk for death, which produced groups of unequal size. Kaplan-Meier survival was assessed for each subgroup. Management of missing values is described in e-Appendix 2.

TABLE 1] Recipient Characteristics by Recipient Age Group

Variable	Age <30 y	Age 30-55 y	Age >55 y
	Mean ± SD or No. (%)		
Age at transplant	24.2 ± 3.2	46.3 ± 7.3	63.4 ± 4.6
Sex (female)	568 (52%)	1,981 (47%)	3,168 (35%)
BMI at transplant (kg/m^2)	19.5 ± 3.36	24.7 ± 4.89	25.9 ± 4.19
Education			
None	2 (0.2)	6 (0.2)	11 (0.1)
Grade school	13 (1.3)	103 (2.7)	254 (3)
High school	428 (42)	1,471 (39)	3,139 (39)
College	370 (36)	1,050 (28)	2,056 (25)
Associate/bachelor degree	175 (17)	853 (23)	1,672 (21)
Graduate degree	37 (3.6)	292 (7.7)	963 (12)
Primary diagnosis			
Group A	51 (4.6)	1075 (26)	3372 (38)
Group B	51 (4.6)	379 (9)	268 (3)
Group C	897 (82)	815 (19)	44 (0.49)
Group D	99 (9)	1932 (46)	5270 (59)
Pretransplant tobacco use	64 (6)	1,963 (48)	6,611 (75)
Chronic steroid use	353 (33)	1,951 (48)	4,083 (47)
Pretransplant functional status (KS)			
<50%	502 (46)	1,469 (35)	2,937 (33)
50%-70%	453 (42)	2,281 (55)	5,122 (58)
>70%	129 (12)	401 (9.7)	814 (9.2)
Insurance			
Medicaid	239 (21.8)	361 (8.6)	290 (3.2)
Medicare	215 (19.6)	1,005 (23.9)	3,912 (43.7)
Private	598 (54.5)	2,644 (62.9)	4,442 (49.6)
Other	46 (4.2)	191 (4.5)	310 (3.5)
ECMO	64 (5.8)	161 (3.8)	110 (1.2)
Mechanical ventilation	146 (13)	337 (8)	452 (5)
Creatinine clearance (mL/min)	113.57 ± 51.7	114 ± 43.2	95 ± 35.6
Bilirubin (mg/dL)	0.53 ± 0.75	0.63 ± 0.85	0.66 ± 0.84
Dialysis	11 (1)	30 (0.72)	13 (0.15)
Cardiac index (L/min/m^2)	3.45 ± 0.98	2.95 ± 0.82	2.79 ± 0.69
Oxygen requirement (in L, at rest)	5.45 ± 5.44	5.72 ± 5.4	5.09 ± 4.93
pCO_2 (mm Hg)	53.4 ± 18.3	49.1 ± 14.7	46.5 ± 12.2
FEV_1%	26.5 ± 15.9	33.9 ± 18.6	42.2 ± 21.4
FVC%	38.9 ± 15.5	44.5 ± 17.1	51.6 ± 17.2
Pretransplant 6MWD (m)	282.9 ± 150.6	248.7 ± 140.5	235.6 ± 134.4
Donor age (y)	31.2 ± 12.7	33.8 ± 13.8	35.2 ± 14.5
Hospitalization in the 90 d before transplant	615 (64)	1313 (37)	1619 (22)
LAS at listing	41.9 ± 14	42.3 ± 15	42.1 ± 14
LAS at transplant	48.9 ± 18	47.9 ± 18	46.2 ± 17
Double lung transplant	1,091 (99.4)	3,566 (85)	4,887 (55)
Single lung transplant	7 (0.6)	635 (15)	5,956 (45)

Age groups were determined on the basis of divergent survival patterns identified in parametric survival model (age <30 y, age 30-55 y, age >55 y). Variables are summarized with number and percentage for categorical variables and mean and SD for continuous variables. LAS is calculated using diagnosis groups: diagnosis group A, obstructive lung disease; diagnosis group B, pulmonary vascular disease; diagnosis group C, cystic fibrosis and immunodeficiency disorders; diagnosis group D, restrictive lung disease. 6MWD = 6-min walk distance; ECMO = extracorporeal membrane oxygenation; KS = Karnofsky Performance Status Score; LAS = lung allocation score.

Results

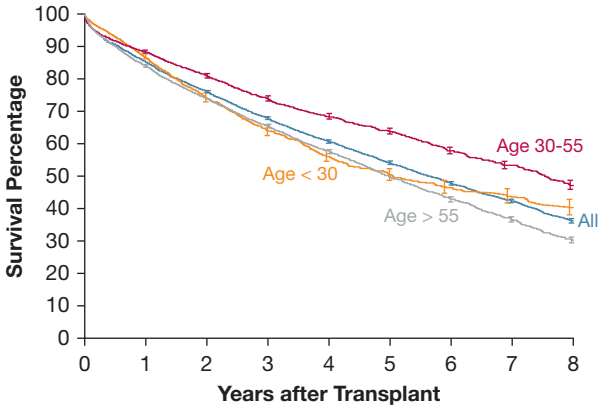
Patient Characteristics

The cohort consisted of 14,253 lung transplant recipients with a median age of 59 years (e-Fig 1). There were 1,098 (7.7%) recipients <30 years, 4,201 (29.5%) 30 to 55 years, and 8,954 (62.8%) >55 years of age. Median posttransplant survival was 5.6 years. A total of 425 patients underwent retransplant. Women recipients composed 52% of those <30, 47% of those 30 to 55, and 35% of those >55 years of age (Table 1). The most prevalent form of insurance in each group was private insurance. Among those with public insurance, recipients <30 years of age had the highest proportion of Medicaid insurance.

The proportion of recipients in each diagnosis group varied by age (e-Fig 2). Recipients in group C were the youngest; those in groups A and D were the oldest. Candidates with idiopathic pulmonary fibrosis received the most transplants (51.2%), followed by those with COPD (31.6%), CF (12.3%), and pulmonary hypertension (4.9%). The recipients in the youngest age group (<30 years) had the greatest level of disability, highest severity of illness, and most frequent utilization of circulatory and ventilatory support (Table 1). Although mean LAS at listing was not statistically different among age groups, mean LAS at transplant was highest in the youngest age group at 49 ± 18.4, and lowest in the oldest age group at 46.2 ± 16.9 (Table 1). The change in LAS from listing to transplant differed among groups, with those <30 years experiencing on average an increase of 7 points, those 30 to 55 years, 5.6 points, and those >55 years, 4.1 points. Younger recipients had greater utilization of pretransplant extracorporeal membrane oxygenation and mechanical ventilation (18.8%) compared with those in mid-age, range 30 to 55 years (11.8%), and those >55 years (6.2%). The youngest age group had the lowest BMI and highest proportion of disability as measured by Karnofsky Performance Status Scale[20] compared with the other age groups.

Association of Survival With Age

Age was a significant risk factor for death at all time points and the magnitude of its effect became more evident as time from transplant increased. Recipients in the youngest and oldest groups had worse survival compared with mid-age patients (Fig 1). The risk-adjusted model for a hypothetical patient with median values for all model variables demonstrated 1-year predicted survival was 93.9% for a 20-year-old, 96.1% for a 45-year-old, and 92.8% for a 65-year-old recipient; 5-year predicted survival for the same age groups was 64.5%, 81.2%, and 71.5%, respectively



Patients at risk

	0	1 y	2 y	3 y	4 y	5 y	6 y	7 y	8 y
All	14,253	10,917	8,145	6,064	4,447	3,110	1,998	1,185	594
<30 y	1,098	870	641	468	321	224	157	93	47
30-55 y	4,201	3,359	2,606	2,002	1,543	1,130	737	476	253
>55 y	8,954	6,688	4,898	3,594	2,583	1,756	1,104	616	294

Figure 1 – *Kaplan Meier survival by recipient age group. Percentage of lung transplant recipients alive by year after transplant. The at-risk table demonstrates the number of lung transplant recipients alive and therefore at risk for the outcome of death, at each time point. Age <30 years (orange), age 30 to 55 years (red), age >55 years (gray), and all patients (blue).*

TABLE 2] Incremental Risk Factors for Death After Lung Transplant

Factor	Coefficient ± SE	*P* Value	R (%)[a]
Early phase			
Older age[b]	1.298 ± 0.17	<.0001	96
Diagnosis group B	0.551 ± 0.13	<.0001	92
Prior cardiac surgery	0.556 ± 0.19	.004	86
Dialysis	1.207 ± 0.25	<.0001	78
Lower albumin	-0.256 ± 0.059	<.0001	85
Higher total bilirubin	0.134 ± 0.019	<.0001	99
Lower cardiac index[c]	0.757 ± 0.014	<.0001	59
Higher mean pulmonary pressure	0.108 ± 0.004	.018	62
Lower % predicted transplant FVC[d]	-0.458 ± 0.12	<.0001	56
Ventilation	0.412 ± 0.11	.0002	94
Functional status disabled (<50%KS)	0.502 ± 0.094	<.0001	81
Nonhospitalized	-0.454 ± 0.11	<.0001	71
Longer waiting list time[e]	0.107 ± 0.023	<.0001	94
Longer ischemic time (h)[f]	0.157 ± 0.044	.0004	95
Higher weight ratio recipient/donor[g]	0.218 ± 0.041	<.0001	89
Earlier year of transplant	-0.118 ± 0.015	<.0001	100
Late phase			
Age			100
Older age[h]	0.767 ± 0.059	<.0001	
Younger age[i]	0.289 ± 0.021	<.0001	
Education lower level	-0.047 ± 0.019	.011	90
CMV IgG	-0.111 ± 0.037	.003	94
Hypertension	0.116 ± 0.048	.15	100
Insurance			99
Medicaid	0.354 ± 0.064	<.0001	
Medicare	0.126 ± 0.038	.001	
Creatinine			78
Higher creatinine[j]	0.301 ± 0.065	<.0001	
Lower creatinine[k]	0.025 ± 0.006	<.0001	
Functional status disabled (<50% KS)	0.156 ± 0.045	.0005	61
Double lung transplant	-0.273 ± 0.038	<.0001	97
Donor older age[l]	0.088 ± 0.041	.031	72
Donor race: black	0.170 ± 0.039	<.0001	81
Donor diabetic	0.196 ± 0.068	.004	77
Donor anti-CMV	0.149 ± 0.039	.0001	94

Multivariable model demonstrating coefficients and reliability (percentage occurrence/1000 bootstrapped model runs) used for creation of the nomogram in Figure 2. Coefficients and SE are included for each variable rather than hazard ratios because of difficulty in interpretation of transformations in hazard ratios. Variables that required multiple transformations because of nonlinearity include separate coefficients, reliability, and *P* values for each transformation. Reliability >50% at a *P* value of .05 suggests a high level of reliability. See Table 1 legend for expansion of abbreviation.

[a]Reliability: % of occurrence of factors in the 1,000 bootstrapped model. See e-Figure 8 for a graphical representation of early and late phase risk.

[b][Log(age)]: logarithmic transformation.

[c][(1/cardiac index)]: inverse transformation.

[d][log(FVC)]: logarithmic transformation.

[e][Log(wait time)]: logarithmic transformation.

[f]$[(\text{Ischemic}/300)]^2$: squared transformation.

[g]$[(\text{Recipient/donor weight ratio})]^2$: squared transformation.

[h]$[(\text{Age}/50)]^2$: squared transformation.

[i]$[(50/\text{age})]^2$: inverse squared transformation.

[j][Log(creatinine)]: logarithmic transformation.

[k]$[(1/\text{creatinine})]^2$: inverse squared transformation.

[l][Log(donor age)]: logarithmic transformation.

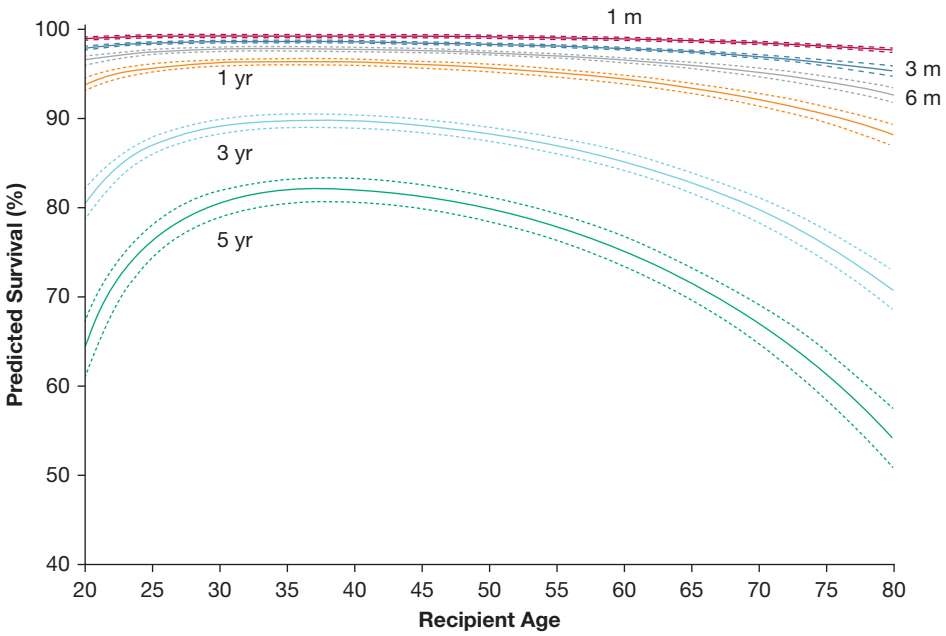


Figure 2 – *Risk-adjusted model of predicted survival by age among lung transplant recipients aged 18 years and older, 2006-2015. Predicted survival by age on the basis of a parametric model adjusting for the variables are included in e-Appendix 1. This model represents a hypothetical patient, of varying ages, with the median values for all variables included in e-Appendix 1. Solid lines represent parametric estimates with a 68% CI represented by dashed lines. The 68% CI was chosen to represent 1 SD width outside the point estimate.*

(Table 2, Fig 2). Five-year survival by parametric (53.4%) and nonparametric methods (53.8%) were similar to observed 5-year survival by the Kaplan-Meier method (54.1%). Both parametric and nonparametric modeling identified that age was the most reliable and important risk factor for posttransplant survival (Fig 2, e-Fig 3). The interaction between age and risk factors for death demonstrated an increase in VIMP across variables, indicating that age improves the predictive ability of most variables (Fig 3). RSF methodology across all ages identified the five most important factors for death by decreasing magnitude as being younger and older in age, higher creatinine, single lung transplant, hospitalization before transplant, and higher total bilirubin (e-Fig 4).

For those aged <30 years, lower education, older donor age, government insurance (Medicare or Medicaid), higher bilirubin, and dialysis had the greatest association with mortality (e-Fig 5). In those aged 30 to 55 years, higher creatinine, diagnosis groups other than CF, pretransplant BMI, recipient race, and donor race were most important (e-Fig 6). In those aged >55 years, higher creatinine, higher bilirubin, hospitalization, lower and higher ratio of donor to recipient predicted total lung capacity, and levels of carbon dioxide had the greatest association with mortality (e-Fig 7).

Discussion

Principal Findings

Among US adult lung transplant recipients, age is the most reliable and important risk factor for death at all timepoints after transplant, with the oldest and youngest experiencing the worst survival. The effect of age on survival becomes stronger as time from transplant increases. Risk factors for death also differ by age; medical factors are important in survival across all age groups, but social determinants of health play a prominent role among those aged <30 years.

Survival and Age in Context

Prior studies addressing mortality among older recipients after lung transplant have focused on the effect of advancing age but included earlier cohorts, defined age as a categorical variable, or reported unadjusted registry data. Similar to our findings, the 2016 International Society for Heart and Lung Transplantation (ISHLT) registry reported that the risk of 1-year posttransplant mortality rose at age 55[21] and multiple studies demonstrated that older recipients, >60 years,[4] >70 years,[8] and >75 years[5] of age had worse survival than younger recipients. In contrast, other studies showed that recipients, >60,[6,7] >65,[7,22] and >70 years of age[9,23] can experience similar survival to those who are younger.

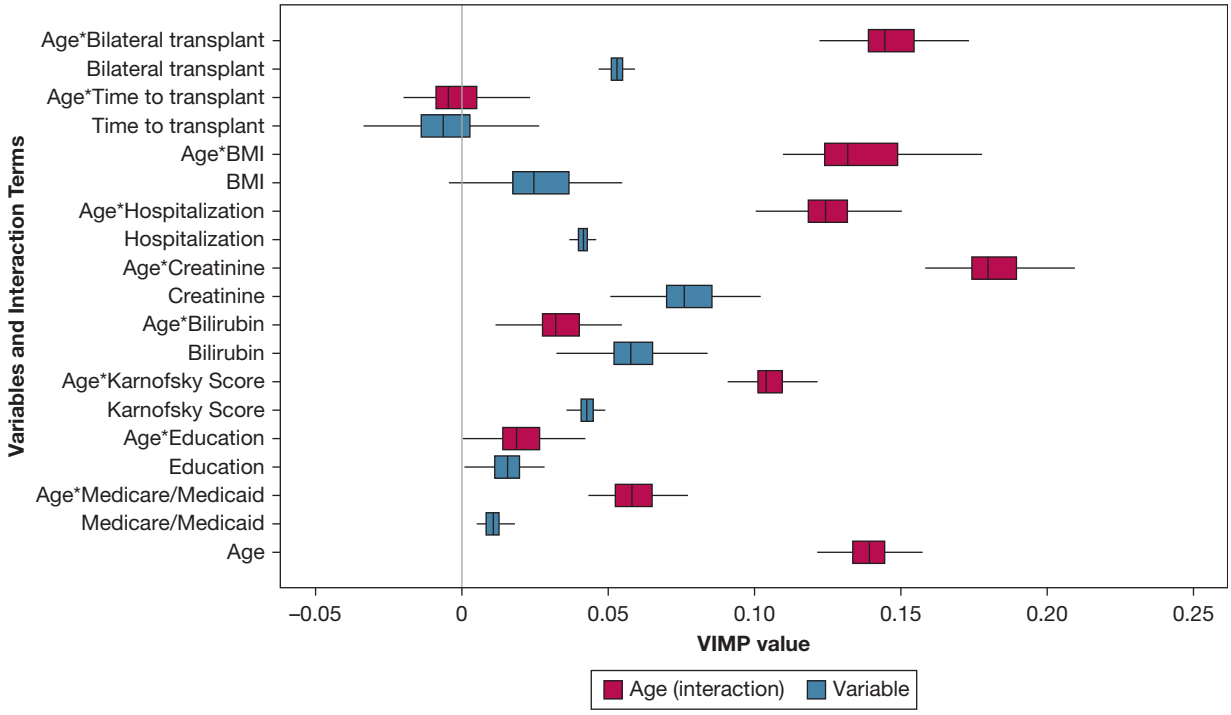


Figure 3 – *VIMP boxplots for main and interaction terms. Interaction terms were analyzed by random survival forest methodology to address interactions with the effect of age. The addition of age increases the VIMP measure for each variable, indicating increased precision of the model for estimating the outcome of death. The VIMP value for age alone is found in the last row of the figure for reference. VIMP = variable importance measure.*

The role of age in survival has been published in the ISHLT annual registry reports,[2] representing a more heterogeneous patient population with data from multiple countries of which the United States comprises only 51%, variable practice patterns, and diverse allocation systems. Our work is the largest study of the effect of age as a continuous variable on survival that uses the US transplant registry including only recipients after the implementation of the LAS system.

Although the finding of decreased survival in older recipients was expected, the observation of decreased survival in younger adult recipients was less expected; however, poor survival in the pediatric and adolescent populations has been previously described. A recent study using the ISHLT registry demonstrated that adolescents aged 14-19 years have worse survival compared to older recipients.[24] These findings are likely multifactorial and factors such as social development, compliance with complex posttransplant medication regimens, and immune responses have been implicated.[24] Stephenson et al[25] demonstrated decreased survival in the youngest and oldest recipients with CF. In our study, this trend of lower survival persisted in younger adults across diagnosis groups, although 82% of younger patients carried a diagnosis of CF.

Important contrasts existed between the youngest and oldest patients at listing and posttransplant. Several listing trends indicate that transplant programs accepted a higher degree of illness as the youngest patients' clinical condition deteriorated while awaiting transplant. The youngest required disproportionately higher use of life support and had greater disability. although their LAS was similar to other age groups at initial listing, their LAS was higher than any cohort at the time of transplant. Not only did those in the youngest age group have the highest LAS at transplant, they also experienced the greatest change in LAS, 7 points, between listing and transplant. Increasing LAS scores during the period on the waiting list have been shown to predict both waitlist mortality and posttransplant survival.[26-29] In a study of the change in LAS, for those with an LAS value >40, as was the case in our population, there was an increasing risk of death on the waiting list with any incremental increase of LAS.[27] Additionally, recipients with an LAS increase of ≥2.5 had an increased risk of posttransplant death providing a possible explanation for our findings of high mortality in our youngest adults.[27] This may reflect a higher tolerance for clinical decline in younger candidates or an ethical imperative to favor younger patients in distribution of limited life-saving societal resources.

Ethical and Policy Implications

Several ethical paradigms can be used to justify preferential access to organ transplant for younger adults. The "fair-innings" perspective provides useful guidance by arguing that every individual deserves to experience a full life and that societal resources should be expended to maximize this chance.[30] The "min-max" perspective requires maximizing the distribution of limited resources to those receiving the least societal advantages. This approach would allow for preferential allocation of organs to younger patients who disproportionately occupy a lower socioeconomic status than their older counterparts and who will invariably experience a premature death. The LAS system, as it applies to adults, stands in contrast to these ethical principles because it prioritizes measures of waitlist mortality, which disproportionately increases access for older adult candidates who inherently have a higher risk of death on the waiting list.

Limitations

This study has limitations intrinsic to a large registry including lack of granularity in variables that may affect survival, missingness, and unmeasured period effects. Although comorbidities such as diabetes, hypertension, and presence of cardiovascular disease were captured in the registry, other conditions such as the presence of gastroesophageal reflux or prior malignancy were not. Another important limitation is that registry data may not capture all aspects of socioeconomic position and is missing data such as median income by ZIP code, employment status, or deprivation resulting from costs related to chronic illness. To circumvent era effect and study a modern cohort, our analysis period began the year after implementation of the US LAS system; however, in a study that spans a 9-year period, there is potential for unmeasured period effects and variability over time. Although the main analysis was performed with age as a continuous variable, age was grouped on the basis of survival models, and these cut-points were not chosen for clinical use or allocation purposes.

Conclusions

These results encompass the lung transplant experience of the United States in the LAS era and represent the largest study of risk factors for death after transplant in the United States. The LAS system changed the behavior of transplant physicians by favoring candidates on the basis of urgency rather than time spent on the waiting list making way for lung transplant recipients that are increasingly older and sicker.[12] We show that among US recipients, the youngest and oldest adults experience the worst survival through divergent pathways that may present opportunities for intervention. This analysis of the US transplant experience can inform the transplant community's efforts to expand recipient access and improve posttransplant survival in future adaptations of the allocation system.

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Author contributions: C. L. and M. V. had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. C. L., E. B., and M. V. take responsibility for study concept and design. L. T. and M. V. acquired the study data. C. L., E. B., L. T., and M. V. analyzed and interpreted the data. C. L., E. B., and M. V. drafted the manuscript with revision of the manuscript for important intellectual content by C. L., E. B., M. V., L. T., K. M., and W. T.

Additional information: The e-Appendixes, e-Table and e-Figures can be found in the Supplemental Materials section of the online article.

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